1. **Scenario: You are working on a project that involves analyzing student performance data for a class of 32 students. The data is stored in a NumPy array named student_scores, where each row represents a student and each column represents a different subject. The subjects are arranged in the following order: Math, Science, English, and History. Your task is to calculate the average score for each subject and identify the subject with the highest average score.**

   Question: How would you use NumPy arrays to calculate the average score for each subject and determine the subject with the highest average score? Assume 4x4 matrix that stores marks of each student in given order.

AIM:

To calculate the average marks for each subject using NumPy arrays and identify the subject with the highest average score.

PROCEDURE:

1. Import the NumPy library.
2. Create a 4×4 array containing student marks in Math, Science, English, and History.
3. Calculate the average of each subject using np.mean() with axis=0.
4. Find the highest average using np.max().
5. Display the average marks and the subject with the highest average.

PROGRAM CODE:

```
import numpy as np
student_scores = np.array([
    [80, 90, 75, 85],
    [70, 85, 80, 75],
    [90, 95, 85, 80],
    [60, 80, 70, 90]
])
subjects = ["Math", "Science", "English", "History"]
average = np.mean(student_scores, axis=0)
print("Average Marks:", average)
highest = np.argmax(average)
print("Highest Average Subject:", subjects[highest])
print("Highest Average:", average[highest])
```

INPUT:

```
[[80, 90, 75, 85],
 [70, 85, 80, 75],
 [90, 95, 85, 80],
 [60, 80, 70, 90]]
```

OUTPUT:

```
average = np.mean(student_scores, axis=0)

print("Average Marks:", average)

highest = np.argmax(average)

print("Highest Average Subject:", subjects[highest])
print("Highest Average:", average[highest])
```

```
Average Marks: [75.  87.5 77.5 82.5]
Highest Average Subject: Science
Highest Average: 87.5
```

RESULT:

Thus, the average marks of each subject were calculated using NumPy. Science has the highest average score of 87.5 marks.